BrightLearn Data Analytics

Research Assignment 1

Foundations of Data Analytics & Data Science

Course	Data Analytics & Data Science
Assignment No.	Research Assignment 1
Target Audience	Beginner Students
Due Date	To be confirmed by instructor
Format	Handwritten answers — scan and email your completed work

Instructions

Read each question carefully and research the answer before writing.

- All answers must be handwritten — typed submissions will not be accepted.
- Write in full sentences and use your own words.
- Include diagrams or examples where relevant — this will earn you bonus marks.
- Cite your sources at the bottom of each section.
- Scan your completed assignment and submit it before the due date.

Section A: Data Fundamentals

Research Tip: *Start with official documentation from platforms like IBM, Google Cloud, AWS, or Microsoft Azure — they offer excellent beginner-friendly explanations.*

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

A Database is just a collection of Information that is stored on a Computer.

- Databases are absolutely essential to the health and success of our country and to the future
- TWO Example where databases are used
1) At Hospistal
2) A BANK.

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse?

- It is a central place where we store multiple database.
i) Regular Database it operate during the day, while Data Warehouse at night it Extract Transactional Load to Data Warehouse at night
ii) To centralize data from multiple database and be able to analyze and ~~make~~ decision-making.

3. What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

- Is a central location that holds a large amount of data in its native, raw format
- Lakes hold raw, unstructured/structured data, while Warehouses hold cleaned, structured data & processed data.
- When a business needing to store massive volumes of raw, unstructured or semi-structured data

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

- Is a focused subset of a data Warehouse tailored to serve the specific needs of a particular business unit, department or user group
- It relate to a data Warehouse because it often extract data from Central DataWarehouse
- A particular group of people in a company, like sales team or finace department

5. Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each.

6. What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.

- Is a conceptual thing on how you want to organize your data
- For efficient data handling ensures that systems can process, store and retrieve information securely.

i) Conceptual models
ii) Logical models.

7. What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action.

- Is a process of finding anomalies, patterns and correlations within large data sets to predicts outcomes.
- In a business you can use the information / outcome to increase revenues
- Cut costs, improve custome relationships, reduces risks and more.
- Example
  i) Retail Market
  ii) Bank.

8. What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

- Is a system used to access the Database and designed to store, retrieve & manipulate data
- MySQL - (Structured Query Language - it is use to manage, manipulate & retrieved data stored in Database Management System
- PostgreSQL - is an free open-source database system that supports both relational (SQL) and non-relational queries, Python & Java
- Oracle - is a proprietary multi-model system used for online transaction processing (OLTP) Data Warehouse and analytics, available on-premises.
- MongoDB - is a document database designed to help developers build modern applications faster.

9. What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics.

- Is a process of combining data from multiple source into a large, central repository called a data Warehouse
  i) It ~~create~~ It clean
  ii) ~~It~~ record It organize raw data
  iii) It update And prepare data for storage & data analytics

So that u can predict the outcome of business decision, generating reports and dashboards.

10. What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

1. CSV – is used for best quick exports and human readability to high- performance
- ~~Hum~~ When to use
- Small files,
- compatibility.
- Quick exports.

It is most used on spreadsheet & it's one of the most straightforward & widely recognized formats for tabular data

2. JSON – It's particularly good at representing data that doesn't fit neatly into simple rows and columns
- API responses,
Configuration files
Storing document-like data in NoSQL database.

3. Parquet – It's a columnar storage format, optimized for efficiency and performance, especially withing a big data ecosystem and it's not designed to be human-readable directly
- It stores data column by column
- The standard format for data storage in data lakes (often stored in object storage like Amazon S3 or google cloud storage feeding data warehouses and efficient processing with frameworks like Apache spark.

4. Avro – is a row-based storage format where data is indexed to improve query performance.
- It defines data types and schemas using JSON Data & stores the data in a binary format (condensed) that help with disk space.
- 10 Mb Avro compressed with SNAPPY algorithm will turn into 6.2 Mb.

Section B: Artificial Intelligence & Machine Learning

Research Tip: *Explore resources like Google's Machine Learning Crash Course, IBM's AI explanations, or OpenAI's blog to build a strong foundational understanding.*

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.

-Is a technology that enables computers and machines to simulate human learning, problem solving, decision-making, creativity and autonomy.
Narrow AI - refers to artificial intelligence systems designed and trained to perform specific tasks or a narrow rage of functions within predefined parameters
General AI - Is a theoretical types of AI capable of performing any intellectual task a human can, with the ability to reason, learn and adapt across unfamiliar domains

12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.

Machine Learning - Is the subset of artificial intelligence focused on algorithms that can learn the patterns of training data and subsequently, make accurate inferences about new data.
The fundamental difference lies in how solutions are created
Supervised - Is a predictive modeling technique where algorithms are trained on labeled datasets containing input-output pairs to learn the maping between them.
Unsupervised - is a type of algorithm that analyzes and clusters unlabeled data to dissover hidden partens, structures or grouping without human interventions
Reinforcement learning - Where an agent learns how to behave in an environment by performing actions and receiving rewards or penalties in return, aiming to maximize the cumulative reward over time.

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?

-Is a specialized subset of machine learning that utilizes multi-layered artificial neural networks to automonomously learn representations of data, mimicking the human brains structure and function
-They can all replicate human intelligence in machines
Neural Networks - are machine learning models inspired by the structure and function of the human brain, designed to recognize patterns & solve complex problems by learning data

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.

- Is a branch of artificial intelligence that enables computers to understand interprete and generate human language in a meaningfull way
- Speech recognations ! Siri virtual assistance like Siri
Alexa and Google Assistance.

Chatbots
Search Engine results
Health Application

15. What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for.

- A large deep-neural-networks that are trained by tens of gigabytes of data that can be used for many tasks.
- It differ from standard machine learning models primary in their learning approach, data requirements and flexibility
- They are used for a wide variety of applications.

16. What is Generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does.

- Is a category of artificial intelligence that uses deep learning models to create original content - such as text, image, audio, video and code - in response to use prompts.
- Chat gpt -
- Gemini
- Github Copilot - is a code completion and programming AI-assistance

17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

- Is the foundational collection of data used to teach a machine learning models how to recognize patterns, make predictions or perform specific tasks by providing examples that the algorithm analyze to adjust it's internal parameters.
- Without quality data, even the most advanced algorithms will fail to produce trustworthy insights or perform reliable in real-world application.
- It's important that your data is cleaned and accurate

18. What is Computer Vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision.

- Is a technology that machines use to automatically reconize images and describe them accurately and efficiently
The machines store images in the form of numbers
1) Drones
2) Retails

Section C: APIs, Tools & Developer Concepts

Research Tip: *Search for 'What is an API for beginners' on YouTube or read documentation from platforms like Postman, RapidAPI, or Anthropic.*

19. What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems?

- Is a set of rules and protocols that enables two sofware programms to communicate and exchange data efficiently without needing to know how the other is built.

1) Weather bureau's

2) Amazon, Agriculture, E-commerce & Manufacturing.

They are important because they are foundational structure of modern software acting as a bridge that enables different applications, systems & databases to communicate and exchange data seamlessly without needing direct access to underlying code & hardware.

20. What is an API Key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly?

- Is a code used to identify an application or user and is used for authentication in computer applications.

It should be stored by a dedicated secret managers like AWS Secrets manager and protected by encrypted files.

- Attackers can impersonate the leginate user or application, leading to unauthorized access to sensitive data, finacial abuse & service disruption.

21. What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs? Briefly describe each method.

- Is a software interface that allows applications to communicate over the internet using HTTP requests while adhering to a set of architectural constraints known as REST principles.
- REST (Representational State Transfer)
- GET - Retriere a resource
- POST - Create a new resource or process data
- PUT - Replace a resource entirely
- DELET - Remove a resource

22. What is JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.

Is a text format that is completely language independent but uses conventions that are familiar to programmers of the C-family of languages, including C, C++, C#

- It is lightweight human-reddable, and language-independent

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23. What is Prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each.

- Is the process where you guide generative AI solutions to generate desired output
- It directly determines the quality, accuracy and relevance of the output.

Poorly written prompt 1) Write a marketing email for my new product
2) Summarize the long document, then write a tweet thread about it, then make a table of all the names mentioned, and fix the grammar in the third paragraph all in a conversional tone.

Better Version 1) Act as a Senior copywriter. Write a 150-word sales emails for a new Eco friendly water bottle targeting environmentally conscious Gen Z professionals.
2

24. What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or JavaScript.

- Is a high-level tool used to transform data by applying a specific operation to every item in a collection.
- It perfoms uniform operations like converting data types e.g. strings to integers, normalizing values or applying mathematics formulas across entire datasets without explicit loops

25. What is Cloud Computing? Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model.

Is the on-demand delivery of computing services, such as servers, storage, databases, networking, software, and analytics, over the internet with a pay-as-you-go pricing model

- IaaS - is a cloud computing model that provides on-demand access to computing resources such as servers, storage, networking, and virtualization.
- PaaS - is a cloud computing model that provide a complete environment for developers to build, deploy, test and manage applications without managing the underlying infrastructure
- SaaS - is a cloud computing model where providers host applications and make them available to users over the internet on a subscription basis.

26. What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are.

- Is a version software tool that tracks and manages changes to files, primarily source code, over time.

Git - is a distributed version control system originally developed in 2005 by Linus Torvalds that tracks changes to files over time, allowing users to revert to previous state compare versions, and manage non-linear workflows.

27. What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists?

- Is an open - source, web-based interactive computing environment that allows users to create and share documents combining live code, equations, narrative text, visualizations, and other rich media.

1) Prython

28. What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name four Python libraries commonly used in data work and briefly describe each.

- Is a computer programming language often used to build websites and software, automate tasks and conduct data analysis.

1) Data analysis and machine learning.

Python has become a staple in data science allowing data analysts and other professionals to use the language to conduct complex statistical calculations, create data visualizations, build machine learning algorithms, manipulate and analyze data and complete other data-related tasks

2) Web development

Is often used to develop the back end of a website or application - that the parts that a user doesn't see.

3) Software testing and prototyping

In software development, Python can aid in tasks like build control, bug tracking and testing. And it can automate testing for new products or features

4. Everyday tasks

- Keep track of stock market or crypto prices
- Send urself text reminders to carry out an umbrella routine when its raining
- Update grocery shopping list
- Fill out online forms automatically

Section D: Data Ethics, Governance & Careers

Research Tip: *Explore the World Economic Forum, GDPR documentation, and LinkedIn Learning for content on data ethics and career pathways in data.*

29. What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to.

- Is the principle that a person should have control over their personal data including deciding how organisations collect store and use it.
- It is critical because it protects individuals from identy theft, Financial fraud and harassment while preserving personal autonomy and civil liberties
- GDPR - is a comprehensive data privacy and security law.
- Austria
- Belgium
- Germany
- Italy and all countries that falls under European Union countries

30. What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic.

- Is a branch of ethics that evaluates and governs the responsible, transperent and fair collections, analysis, sharing and use of data throughout it's entire lifecycle and practices that have the potential to adversely impact people and society

1) Clearview AI, which scraped billions of personal photos from social media to build facial recognition databases without consent, resulting in a $51,75 million settlement in 2025 for violating privacy law

31. What is Data Governance? Define it and explain the key components of a good data governance framework.

- Is a comprehensive approach that comprises the principles, practices and tools to manage an organizations data's data asset throughout their lifecycle.

1) Data quality management - establishes process for validation, cleansing and profilling to to ensure data is accurate, complete and reliable.

2

32. What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences.

- Is the system tendency of an AI system to produce outputs that unfairly favor or disadvantage certain groups or outcomes

Root causes, Bias AI typically stems from incomplete or skewed training data that does not represent the full population, historical prejudices reflected in past records, or flawed algorithmic design that prioritizes certain features over others

Real-world impact: Baised AI can lead to unfair hiring decisions, discriminatory loan denials, lower diagnostic accuracy for minority groups in healthcare & reinforced racial profiling in predictive policing

33. What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills, and tools.

ROLE	PRIMARY FOCUS	KEY SKILLS	COMMON TOOLS
DATA Analyst	Deriving insights and trends from processed data to supports business decision-making	SQL, statical analysis, data visualization, business communication	Excel, Tableau, Power BI, SQL, Python (PANDAS
DATA ~~SCIENCE~~ ENGINEER	Building and maintaining scalable data infrastructure and pipeline	Data Architecture, ETL/ELT Process, cloud Platforms, big data framework.	Apache spark, SQL Kafka, Airflow AWS Python, JAVA
DATA ~~ENGINEER~~ SCIENTIST	~~Building and maintaining scalable infrastructure~~ Analizing data using advanced algorithms and statistics to predict outcome	Machine learning, statical modeling, predictive analytic, data story-telling	Python, R, Tensorflow Pytoch, Scikit-learn, SQL Tableau
AI ENGINEER	Building and deploying AI-Powered applications and intelligent systems	Software engineering, LLM APIS, prompt engineering RAG Systems deployment	python, Lang chain Vector Databases REST APIS, Docker Kubernetes

34. What is Business Intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions.

Is a technology-driven process that collects, analyzes, and visualizes raw data from various internal and external sources to provide actionable insights for strategical and tactical decision-making

Power BI - is a platform that connects to diverse data sources, transform that connects to diverse data sources, trans and models that data, and visualizes it into interactive reports and dashboards.

Section E: Personal Reflection

Answer the following reflection questions thoughtfully and honestly. There is no right or wrong answer — we want to know what you think!

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

DATA ENGINEERING

36. Which concept did you find most difficult to understand? What resources did you use to help you understand it better?

AI ENGINEERING, I fail to understand why human being should be replaced by machine

37. How do you see Data and AI affecting your future career or everyday life? Be specific with examples.

They are fundamentally shifting careers from routine execution to strategic oversight and complex problem solving, while simultaneously transforming daily life through personalized automation and enhanced decision-making. Impact on everyday life- AI enhances efficiency and safety by automating mundane chores, managing complex data for health diagnostics and powering personalized recommendations in entertainment and shopping.

References

List all the sources you used to complete this assignment. Use the format below:

Author/Organisation. (Year). Title of article or webpage. Website Name. URL

1. SAS Institute (2026) About Data mining History. www.sas.com
2. Robert Agar/ER-Studio (2026) DATA MODELS. www.erstudio.com
3. W3School. 1996-2026. About Postgres SQL, My SQL, Oracle, Mongo DB
4. AWS / (2026) About ETL. www.amazon.com
5. Apxml Machine learning (2026) Common Data Storage used for analytics. www.apxml.com
6.
7.
8.

Good luck! We believe in you.
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